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27 June 1975
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XWS-6302
24 August 1966

MILITARY SPECIFICATION

CABLE, TORPEDO CONTROL, TYPE "B", MK 7 MODS 0, 1, AND 2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers requirements for the procurement of torpedo control cable assemblies, Type "B", in accordance with LD 162531, LD 162533, and DL 2865325.

2 APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-621	Boxes, Wood, Nailed
PPP-C-843	Cushioning Material, Cellulosic

Military

MIL-B-117	Bags, Sleeves, and Tubing; Interior Packaging
MIL-S-901	Shock Tests, H. I. (High Impact): Shipboard Machinery, Equipment, and Systems; Requirements for
MIL-C-82630	Cable, Electric, 65-Conductor

STANDARDS

Federal

Federal Test Method Standard No. 228	Cable and Wire, Insulated, Methods of Testing

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners or Pallet-type Base
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment

. DRAWINGS

Naval Sea Systems Command

LD 162531	Cable, Torpedo Control, Mk 7 Mod 0
LD 162533	Cable, Torpedo Control, Mk 7 Mod 1
DL 2865325	Cable, Torpedo Control, Mk 7 Mod 2

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LD 162288	Connector Receptacle
LD 162838	Connector, Cable, Tube Door, Mk 42 Mod 1
Dwg 757989	Fiberboard Container
Dwg 2865910-3	Connector Plug, Electrical

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3 REQUIREMENTS

3.1 Description. The torpedo control cable assembly consists of a tube door plug, a 65-conductor electric cable, and a connector plug.

3.2 General requirements. The torpedo control cable assembly shall be in accordance with LD 162531 for the Mk 7 Mod 0, with LD 162533 for the Mk 7 Mod 1, or with DL 2865325 for the Mk 7 Mod 2; as specified in the contract or order. See 6.2.1.

3.3 Classification of requirements. The requirements for the cable assemblies are classified herein as follows:

Requirement	Paragraph
Preproduction	3.4
Materials	3.5
Design and Construction	3.6
Mechanical Requirements	3.7
Electrical Requirements	3.8
Environmental Requirements	3.9

3,4 Preproduction sample. Unless otherwise specified in the contract or order, a preproduction sample of the cable shall be manufactured using the methods and procedures proposed for the production lot. This sample will be tested as specified in 4.3.1 and is for the purpose of determining, prior to starting production, that the manufacturer's production process will produce cables that will meet the requirements of this specification.

3.5 Materials. All materials shall be as specified herein and in accordance with applicable drawings.

3.6 Design and construction. The design and construction shall be as specified herein and in accordance with applicable drawings.

3.6.1 Tube door plug. The tube door plug shall consist of a machined aluminum casting, a polycarbonate insulator, and copper alloy contacts. It shall mate with a Torpedo Tube Cable Connector Mk 42 Mod 1 in accordance with LD 162838. The contacts shall be crimped to the 65-conductor cable conductors. A trigger-type latch shall keep the plug properly mated.

3.6.2 Cable. The 65-conductor cable shall be in accordance with MIL-C-82630.

3.6.3 Connector plug. The connector plug shall consist of a threaded coupling, an aluminum shel a polycarbonate insulator, and copper alloy contacts. The connector plug of cable assembly Mk 7 Mo 0 or 1 shall mate with a B-C Adapter in accordance with LD 162288. The connector plug of cable assembly Mk 7 Mod 2 shall be in accordance with Dwg 2865910-3 and shall mate with the "C" cable connector of SSN 688 class submarines. The contacts shall be crimped to the 65-conductor cable conductors.

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3.7 Mechanical requirements. The cable assembly shall successfully meet the following mechanical requirements:

3.7.1 Mating or unmating force. The tube door plug shall fully engage or disengage with an axial force of not more than 60 pounds (267 N). See 6.3.

3.7.2 Cable strength. The cable assembly shall withstand a 200-pound (890-N) load applied in accordance with 4.5.3 without functional failure,

3.7.3 Cable flexibility. The cable assembly shall withstand the torsion and low temperature bending specified in 4.5.4 without functional failure.

3.8 Electrical requirements. The cable assembly shall successfully meet the following electrical requirements:

3.8.1 Dielectric strength. The cable assembly shall withstand, without arcing, a 5-second application of 1000 volts as to any conductor with all adjacent conductors and shields at ground potential.

3.8.2 Insulation resistance. The insulation resistance between any conductor or shield of the cable assembly and all others connected in parallel shall be greater than 70 megohms when measured with a potential of 500-750 vdc.

3.8.3 Continuity. The cable assembly shall maintain continuity between each pair of corresponding connector plug and tube door plug contacts.

3.8.4 Contact, conductor resistance. The maximum resistance of each circuit shall be as shown in Table 1.

TABLE I. CIRCUIT RESISTANCE

Conductor AWG Size	Maximum DC Resistance
22	0.6 ohm
18	0.4 ohm
16	0.4 ohm

3.9 Environmental requirements. The cable assembly shall be capable of meeting the other requirements herein during or after being subjected to the following environmental conditions.

3.9.1 Vibration. The cable assembly shall successfully withstand a vibration test in accordance with MIL-STD-167, Type I. During this vibration there shall be no circuit discontinuities exceeding 100 microseconds in duration.

3.9.2 Shock. The cable assembly, mounted to simulate service installation, shell withstand the shock tests of MIL-S-901 for lightweight equipment.

3.9.3 Salt water immersion. The cable assembly, with connector attached, shall successfully withstand immersion in salt water for 30 16-hour periods with 8 hours between immersions.

3.9.4 Low temperature (operating). The cable assembly shall withstand exposure to 23″ F (-5° C) for 48 hours.

3.9.5 High temperature (operating). The cable assembly shall withstand exposure to 135° F (57° C) for 48 hours.

3.9.6 Temperature and humidity (operating). The cable assembly shall withstand the temperature and humidity (operating) test of 4.5.6.6.

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3.9.7 Temperature and humidity (storage mode). The cable assembly s-hall withstand the temperature and humidity test of 4..6.7.

3.9.8 Tropical environment (fungus). The cable assembly shall withstand exposure for 28 days to 86° F (30° C), 90 percent relative humidity, and fungus without showing evidence of being a fungus nutrient.

3.10 Workmanship. The cable assemblies furnished to this specification shall be constructed and finished in a manner to assure full compliance with all requirements of this specification and applicable NAVSEA drawings. Particular attention shall be paid to the neatness and quality of soldering, wiring, and freedom from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements relative to workmanship insofar as not specifically covered by applicable specifications.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified, unless disapproved by the Government, The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4,1.1 Mating connectors and cables required for sampling inspections will be furnished by the procuring activity.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. Preproduction inspection. (See 4.3.1,)
2. Acceptance inspection. (See 4.3.2.)

4.3 Inspections.

4.3.1 Preproduction Inspection.

4.3.1.1 Preproduction samples. Four of each type of cable assembly shall be manufactured by the contractor using the same procedures and methods and at the same plant location proposed by the contractor for the-execution of the contract. The samples shall be tested for the purpose of determining compliance with the requirements of the contract or order and the applicable specifications. Further production of the item by the contractor prior to the approval of the procuring activity or completion of inspection and tests on the preproduction sample shall be at the contractor's risk.

4.3.1.2 Preproduction inspection procedure. Preproduction inspection shall be performed as specified and in the sequence shown in Table II. When specified in the contract or purchase order, certified inspection data covering the results of preproduction inspections shall be forwarded to the procuring activity. See 6.2.2.

4.3.1.3 Failure of sample. The failure of any preproduction sample to comply with any provision of this specification shall be cause for denial of production approval. The disposition of accepted and rejected preproductin samples, and corrective action shall be in accordance with the contract or purchase order, See 6.2.1.

4.3.2 Acceptace inspection.

4.3.2.1 Every cable assembly submitted for acceptance shall be subjected to acceptance inspection.

4,3.2.2 Acceptance inspection procedure. Acceptance inspection shall be performed as specified and in the sequence shown in Table 111,

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TABLE II. PREPRODUCTION INSPECTION

Examination or Test	Requirement Paragraph	Method Paragraph
Acceptance Inspection		4.3.2
Cable Strength	3.7.2	4.5.3
Cable Flexibility	3.7.3	4.5.4
Dielectric Strength	3.8.1	4.5.5.1
Vibration	3.9.1	4.5.6.1
Insulation Resistance	3.8.2	4.5.5.2
Continuity	3.8.3	4.5.5.3
Shock	3.9.2	4.5.6.2
Insulation Resistance	3.8.2	4.5.5.2
Continuity	3.8.3	4.5.5.3
Salt Water Immersion	3.9.3	4.5.6.3
Insulation Resistance	3.8.2	4.5.5.2
Continuity	3.8.3	4.5.5.3
Low Temperature (Operating)	3.9.4	4.5.6.4
High Temperature (Operating)	3.9.5	4.5.6.5
Temperature and Humidity (Operating)	3.9.6	4.5.6.6
Insulation Resistance	3.8.2	4.5.5.2
Mating and Unmating Force	3.7.1	4.5.2
Temperature/Humidity (Storage Mode)	3.9.7	4.5.6.7
Tropical Environment (Fungus)	3.9.8	4.5.6.8

TABLE III. ACCEPTANCE INSPECTION

Examination or Test	Requirement Paragraph	Method Paragraph
Examination of Product	3.2, 3.5, 3.6, 3.10, 5	4.5.1
Insulation Resistance	3.8.2	4.5.5.2
Connector Mating and Unmating Force	3.7.1	4.5.2
Contact, Conductor Resistance	3.8.4	4.5.5.4

4.3.2.3 Failure. The failure of a cable assembly to comply with any provision of this specification shall be cause for rejection of that cable assembly and may be cause for revocation of production approval. The disposition of rejected cable assemblies and corrective action shall be in accordance with the contract or purchase order. See 6.2.1,

4.4 Test conditions. Unless otherwise specified, tests and examinations required by this

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specification shall be conducted under any combination of conditions within the following changes,

(a) Temperature: 20 degrees C to 30 degrees C.

(b) Relative humidity: Room ambient to 95 percent.

(c) Barometric pressure: 24 to 31 inches of mercury (80 to 105 kPa).

4.5 Test methods

4.5.1 Examination of product. The cable assemblies-and components shall be examined to insure conformance to the following requirements of this specification, applicable NAVSEA drawings, and the contract or order.

(a) 3.2 NAVSEA drawings,

(b) 3.5 Materials,

(c) 3.6 Design and construction.

(d) 3.10 Workmanship.

(e) 5 Preparation for Delivery,

4.5.2 Connector mating and unmating force. A Torpedo Tube Cable Connector Mk 42 Mod 1 shall be mounted in a movable holding device capable of measuring force in both a push and pull direction. The appropriate cable assembly tube door plug shall be engaged and disengaged to the mating con-connector by hand as in service use. The force required to engage and disengage the connector shall be determined. See Figure 1.

4.5.3 Cable strength. Securely mount a Torpedo Tube Cable Connector Mk 42 Mod 1 in a suitable fixture (Figure 2). Mate the appropriate cable assembly tube door plug with the connector and se-curely clamp the cable in a horizontal plane 15 +1 -O inches (380+25-Omm) from the connector. Attach the weight support to the cable at a point midway between the plug and cable clamp. The weight sup-port shall be no more than 1.5 inches (38mm) long and of such design as to spread the load evenly over the 1.5-inch (38-mm) length of cable. The cable sheath may be protected or padded to prevent damage, A weight of 200 pounds (890N) shall be suspended from the cable for a period of at least 60 seconds, not including the application and removal periods. During application and removal of this force, all circuits shall be monitored for a discontinuity of greater than 100 microseconds. The circuits may be wired in series. The load shall be applied and released a total of 20 times over a period not to exceed 2 hours.

.4.5.4 Cable flexibility

4.5.4.1 Torsion test

(a) A torsion test shall be performed on the cable assembly while it is mounted in a suitable test fixture. The cable shall be mounted and clamped at two points on the test fixture such that a 90-degree bend with a radius not to exceed 4 inches (102mm) shall be formed within a 10-inch (254mm) length of cable (see Figure 3). The cable shall be rigidly secured to a fixed support at one clamp position. The pivot axis of the arm shall be an extension of the centerline of the cable which is clamped to the fixed support.

(b) One cycle of torsion shall consist of a 90-degree rotation in one direction and a 180-degree rotation in the reverse direction, and a 90-degree rotation in the original direction to the start position, in a plane perpendicular to the fixed support.

(c) The test shall be conducted at a rate of 2 cycles per minute. The time duration of one cycle shall approximate but not exceed 15 seconds. Continuity of all conductors shall be monitored continuously. The test shall be stopped and the number of cycles recorded in the event of failure of any conductor. The test shall consist of 3000 cycles of torsion.

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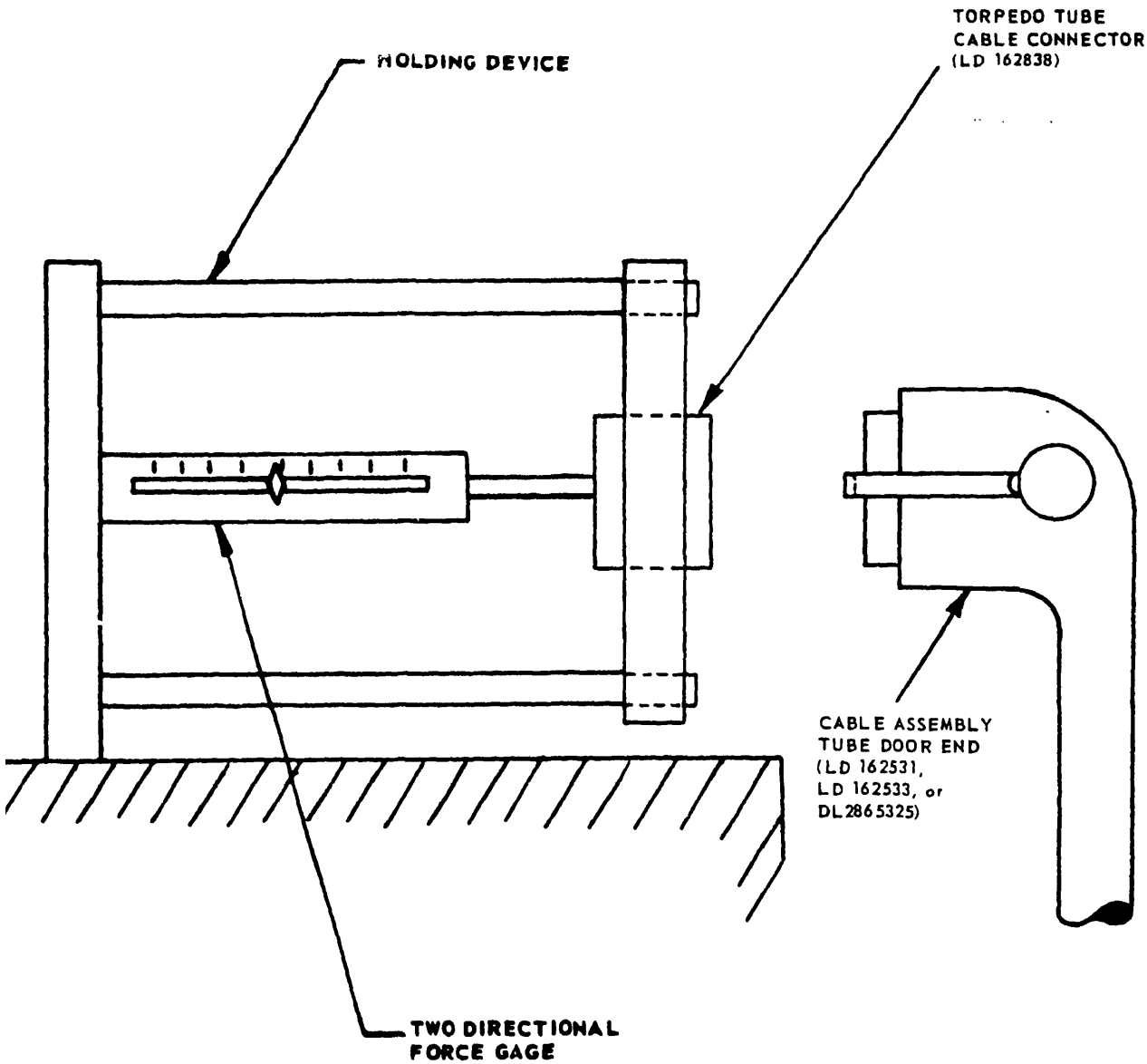


FIGURE 1. TEST SETUP - MATING AND UNMATING FORCE.

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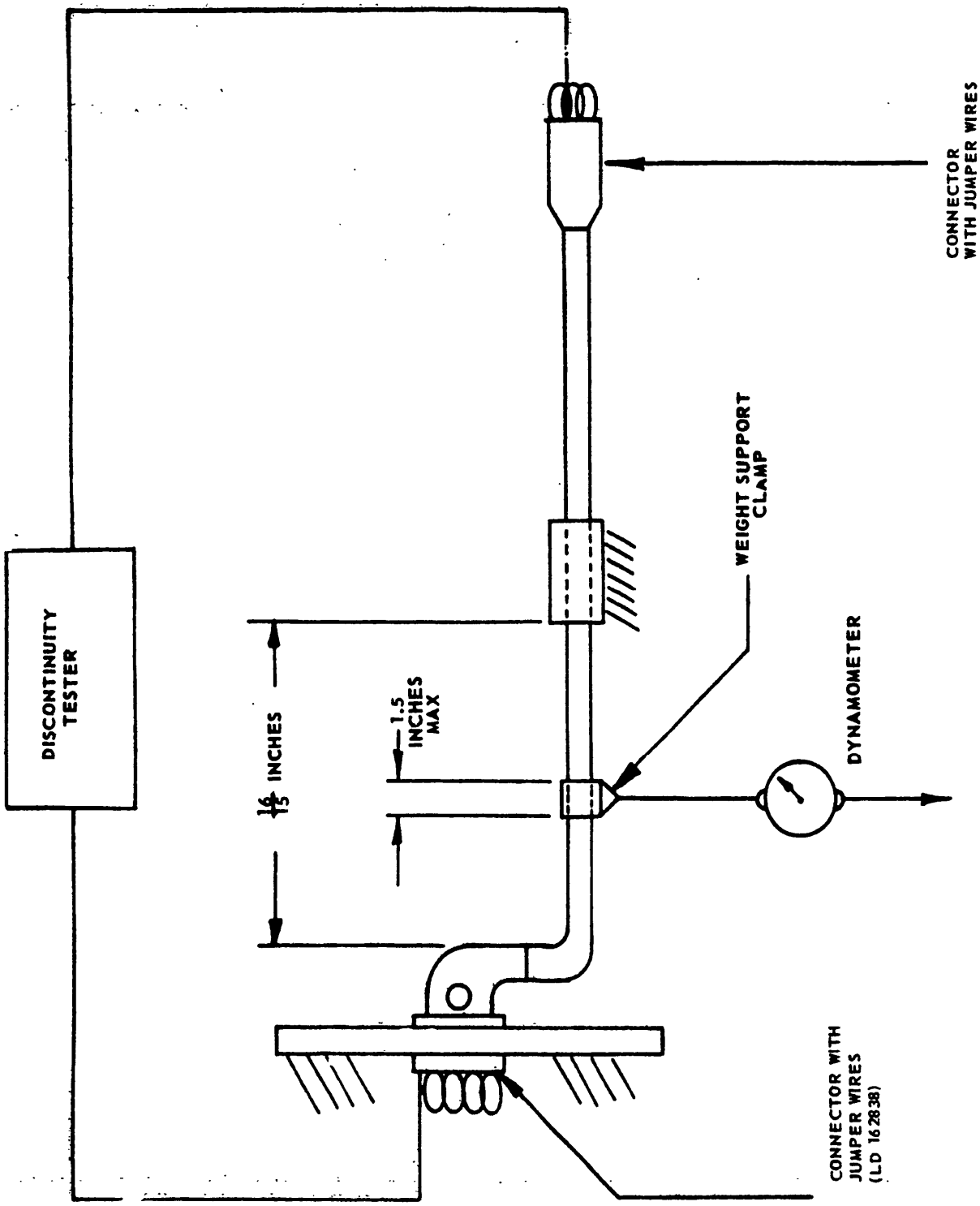


FIGURE 2. TEST SETUP – CABLE STRENGTH.

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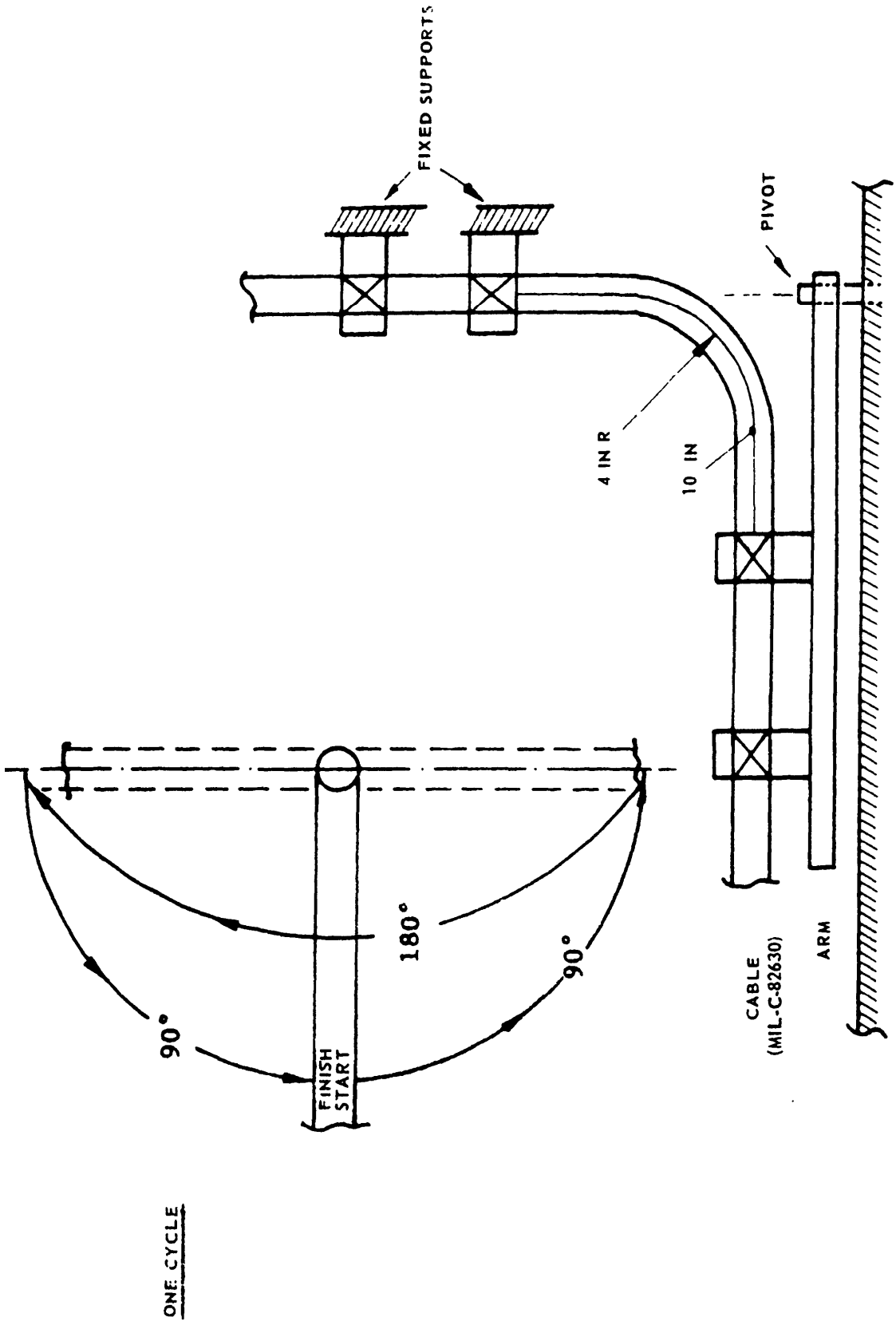


FIGURE 3. TORSION TEST SETUP.

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4.5.4.2 Low temperature (90-degree bend test)

(a) While the cable is mounted in a suitable fixture, a 90-degree bend test shall be performed at -40 ±3 degrees F (-40°±2° C). The cable assembly and test fixture shall be conditioned at the test temperature for at least 12 hours prior to initiation of the test.

(b) The cable assembly shall be mounted and clamped at two points on the test fixture such that a 90-degree bend with a radius not to exceed 6 inches (152mm) shall be formed within a 12-inch (305-mm) length of cable (see Figure 4). The cable shall be rigidly secured to the underside of a fixed support lying in a horizontal plane at one clamp position, and rigidly secured to a pivoting arm which shall be so mounted as to oscillate from a vertical position to a horizontal position. When in its horizontal position, its axis shall form and extension of the plane of the fixed support, and the cable shall be attached to the underside of the arm. The attachment of the cable to the oscillating arm shall be made when the arm is in the vertical position. The pivot axis of the oscillating arm shall intersect the longitudinal axis of the fixed support and shall be equidistant from the two clamping points. The fixture will be so designed that the section of cable between the clamp positions shall not touch any portion of the mechanism or cold box during the test.

(c) One cycle of test shall consist of an oscillation of the pivot arm from the vertical position to the horizontal and back to the vertical. The test shall be conducted at a rate of one cycle per minute, with each half cycle to be accomplished within 6 seconds. The delay between cycles may occur at the end of either half cycle or may be divided to occur at the end of each half cycle.

(d) Continuity of all conductors shall be monitored continuously. The test shall be stopped and the number of cycles recorded. in the event of failure of any conductor. The test shall consist of 500 cycles of bending.

4.5.5 Electrical tests

4.5.5.1 Dielectric strength. The cable assembly shall be tested dry, without immersion in water, in accordance with Federal Test Method Standard No. 228 (Method 6111) to determine conformance with 3.8,1.

4.5.5.2 Insulation resistance. The cable assembly shall be tested dry, without Immersion in water, in accordance with Federal Test Method Standard No. 228 (Method 6031) to determine conformance with 3.8.2.

4.5.5.3 Continuity. Each circuit of the cable shall be probed with a 6-12 vdc buzzer, or equivalent, for continuity.

4.5.5.4 Contact, conductor resistance. The cable assembly shall be mated to a Torpedo Tube Connector Mk 42 Mod 1 and the resistance measurement made from each female socket of the connector to the corresponding male pin in the in-line connector (see Figure 5). The ohmmeter or Kelvin bridge shall be capable of measuring a resistance of less than 0.1 ohm with an accuracy of at least ±0.02 ohm. The resistance of each circuit shall be determined.

4.5.6 Environmental tests

4.5.6.1 Vibration. The vibration test shall be performed with the cable in the mated condition. The mated cable assembly and torpedo tube cable connector shall be mounted to a suitable vibration fixture and this fixture in turn shall be mounted to a sinusoidal vibration system as illustrated in Figure 6. The vibration fixture shall be sufficiently rigid to prevent resonant frequencies greater than 1.5 to 1. All circuits of the mating connectors shall be wired in series. The input control accelerometer shall be placed on the vibration fixture adjacent to the cable tube door mounting point. Two monitoring accelerometers shall be mounted to the cable assembly, one on the in-line connector end and one on the tube door end.

4.5.6.1.1 Vibration schedule and levels. Vibration shall be conducted in three mutually perpendicular directions (see Figure 6) in accordance with the following schedule. All tests. in one direction shall be completed before proceeding to tests in another direction.

(a) Resonance search. The sample shall be vibrated at frequencies from not higher than 10 Hz to not less than 33 Hz at a table vibratory amplitude of 0.010 ±0.002 inch (0.25±0.05mm) and from not

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more than 34 Hz to not less than 50 Hz at a table vibratory amplitude of 0.003 +0,000 -0.001 inch (0,075 +0.000 -0.025 mm). The change in frequency shall be made in l-Hz ±0.1 Hz steps and maintained at each step for 15 ±5 seconds. The frequencies at which the greatest resonance or discontinuities in excess of 100 microseconds occur shall be noted.

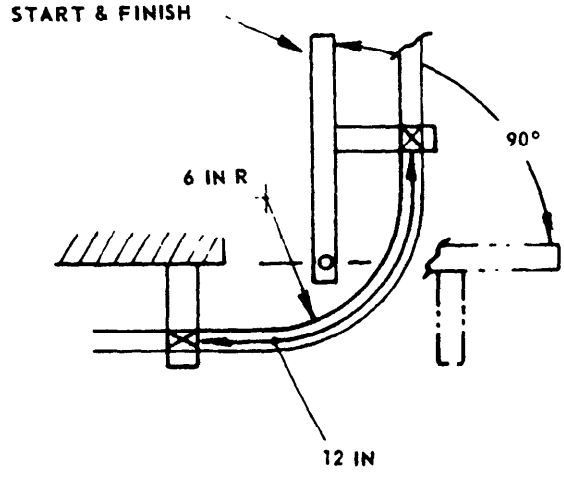


FIGURE 4. TEST SETUP – 90-DEGREE BEND LOW TEMPERATURE.

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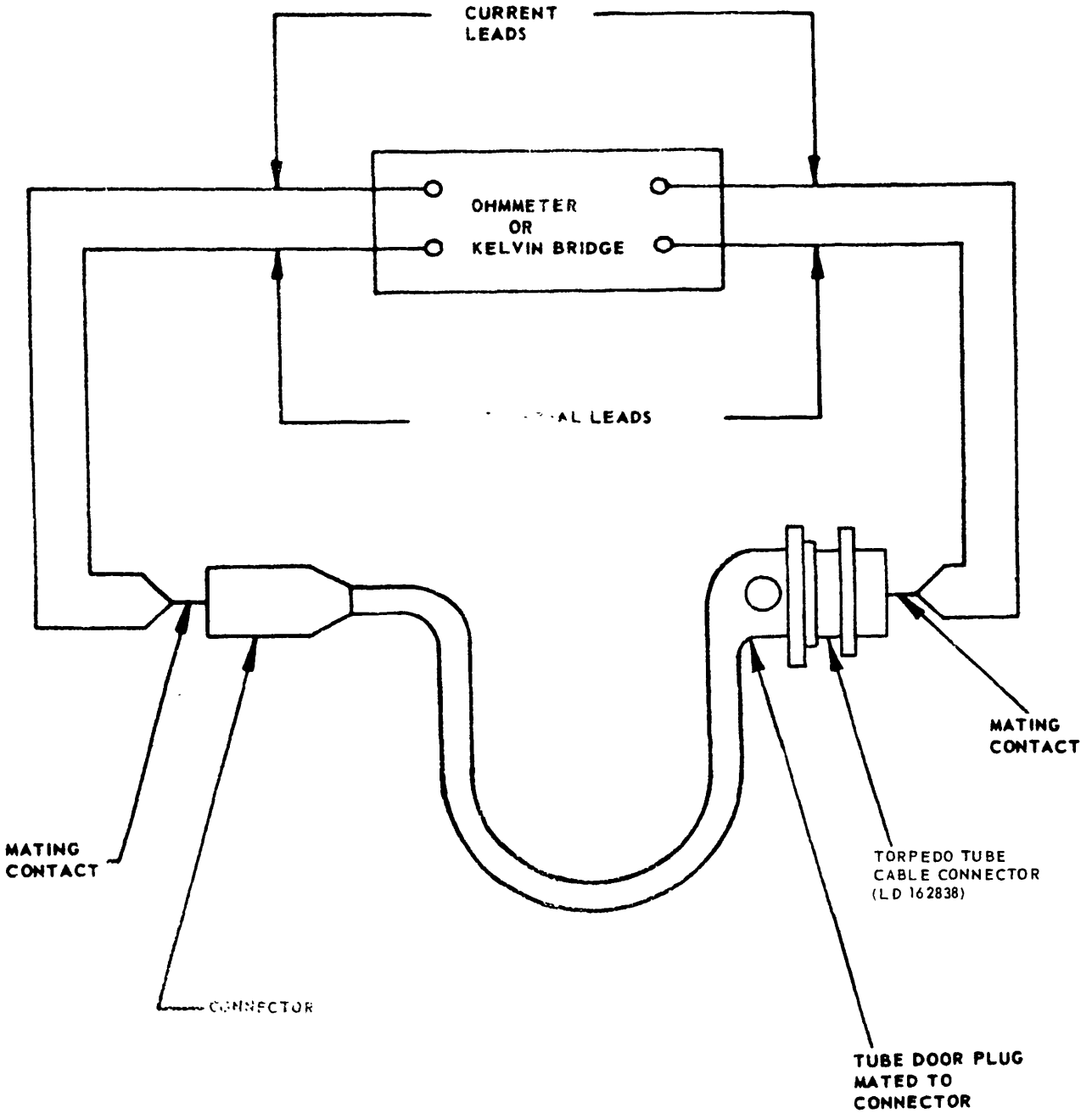


FIGURE 5. TEST SETUP – CIRCUIT RESISTANCE.

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(b) Vibration Levels. The sample shall be vibrated in accordance with Table IV. The dwell time at each integral frequency shall be not less than 5 minutes.

TABLE IV. Vibration Levels

Frequency Range (Hz) (Nominal)	Table Amplitude (See Note Below)	
	Inches	mm
4 to 15	0.030 ± 0.006	0.75 ± 0.15
16 to 25	0.020 ± 0.004	0.50 ± 0.10
26 to 33	0.010 ± 0.002	0.25 ± 0.05
34 to 40	0.005 ± 0.001	0.125 ± 0.025
41 to 50	0.003 + 0.000 - 0.001	0.075 + 0.00 - 0.025

NOTE : All references to amplitude of vibration refer to maximum displacement of sinusoidal motion from the position of test, or one half the total excursion,

(c) Resonance dwell. The sample shall be vibrated for not less than 2 hours at the frequency at which the greatest resonance or discontinuities were noted. If no discontinuity or resonance occurs, resonance dwell shall be performed at 50 Hz.

4.5.6.1.2 The above schedule shall be completed in each of the three mutually perpendicular axes.

4.5.6.2 Shock. The cable assembly shall be mated in Torpedo Tube Cable Connector Mk 42 Mod 1 and securely mounted to a shock fixture simulating service conditions as shown in Figure 6. All circuits shall be wired in series for the purpose of monitoring any discontinuities. The test specimens shall be subjected to the shock test defined In MIL-S-901 for lightweight equipment. Three shocks shall be applied parallel to each of the three principal axes as shown in Figure 6. A total of nine shocks shall be applied. During shock, the series circuit shall be monitored for discontinuity, using a relay chatter indicator or equivalent. Discontinuities in excess of 100 microseconds shall be considered evidence of failure.

4.5.6.3 Salt water immersion. The cable assembly, mated with Torpedo Tube Door Connector Mk 42 Mod 1, shall be subjected to a 30-day immersion test. The cable assembly shall be attached to a fixture so that all of the plug end (tube door end) will be immersed in sea water. The cable assembly shall be immersed for not less than 16 hours and removed for a period of not less than 8. hours. The sequence shall be repeated 30 times. No fresh water rinse will be permitted during the 30 cycles.

4.5.6.4 Low temperature (operating). The cable assembly shall be mated to Torpedo Tube Cable Connector Mk 42 Mod 1 and mounted in the test chamber. The chamber temperature shall be lowered to +28 degrees F ±5 degrees F (-2°±3°C) and maintained for a period not less than 48 hours. At the end of this time and while still at the test temperature, the test specimen shall be subjected to the tests of 4.5.5.2, 4,5.5.3, and 4.5.2.

4.5.6.5 High temperature (operating). The cable assembly shall be mated to Torpedo Tube Cable Connector Mk 42 Mod 1 and mounted in the test chamber. The chamber temperature shall be raised to 130 degrees F ±5 degrees F (54°±3°C) and maintained for a period not less than 48 hours. At the end of this time and while still at the test temperature, the test specimen shall be subjected to the tests of 4.5.2, 4.5.5.2, and 4.5.5.3.

4.5.6.6 Temperature and humidity (operating). The test unit shall consist of the mated assembly of the "B" torpedo control cable Mk 7, one cable connector Mk 42 Mod 1, and one "A" cable

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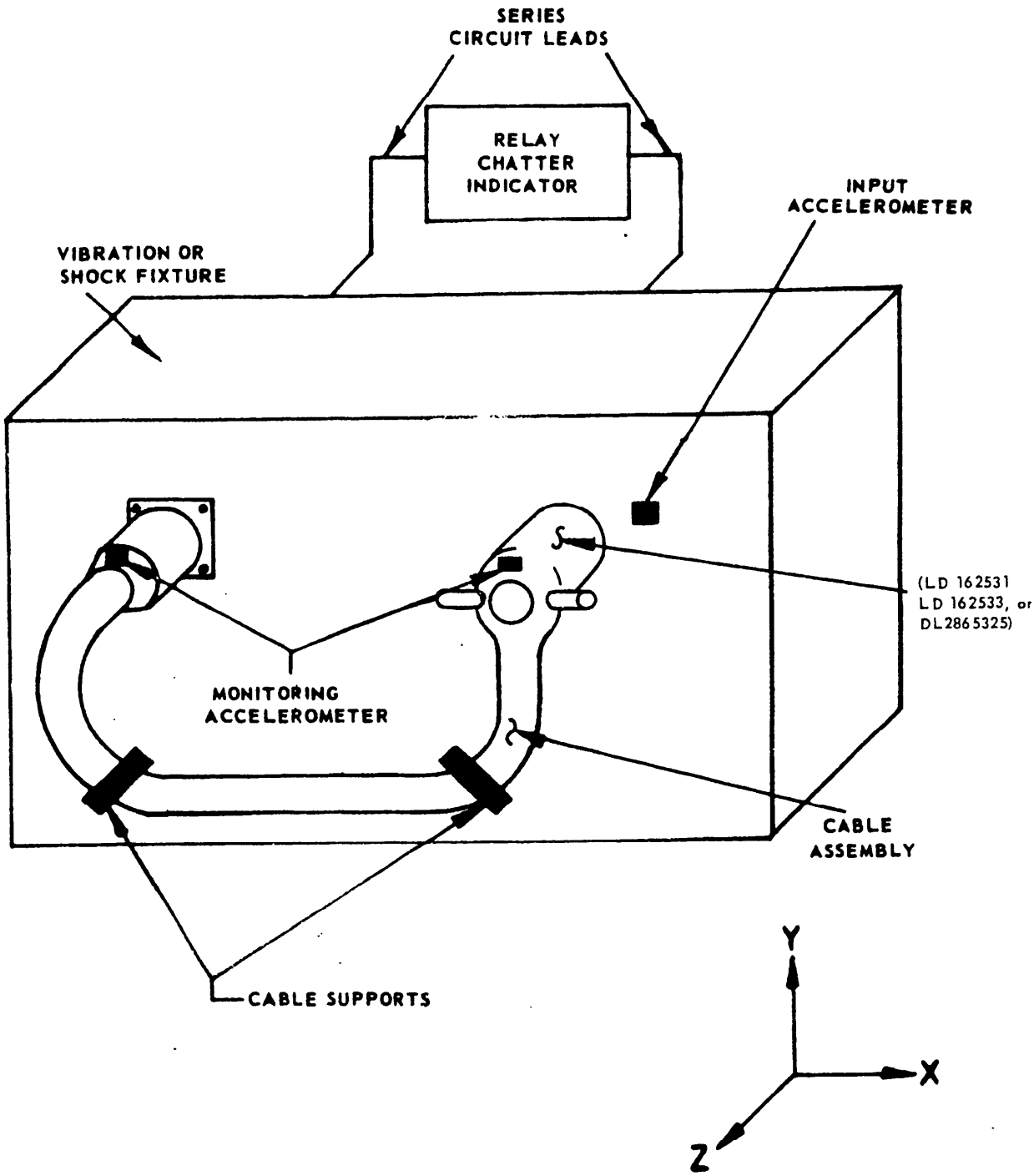


FIGURE 6. TEST SETUP - VIBRATION AND SHOCK.

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Mk6 Mod 0. The prepared unit shall be placed in a test chamber and subjected to a temperature of +233 degrees F ±5°F (50°±3°C) for not less than 2 hours, During this 2-hour period the tests of 4.5.5.2 and 4.5.5.3 shall be performed.

(a) After 2 hours, the test units shall be subjected to five 24-hour cycles of not less than 16 hours at +122 degrees F ±5°F (50°±3°C) and 8 hours at +86 degrees F ±5°F (30°±3°C). The relative humidity shall be maintained constant at not less than 95 percent during the five cycles. During each cycle and while at +122 degrees F (50°C) the tests of 4.5.5.2 and 4.5.5,3 shall be performed.

(b) Upon completion of five cycles, the temperature shall be maintained at +122 degrees F ±5°F (50°±3°C) for not less than 8 hours. During the 8-hour period, the relative humidity shall be 50 ±5 percent for the first 2 hours, 95 percent during the next 4 hours, and 40 percent during the remaining time. At 30-minute intervals ±5 minutes, the test of 4.5.5,3 shall be performed.

NOTE : The transitions between temperatures shall be accomplished within the 8-hour period so that the time at the high temperature is approximately 16 hours. Each transition shall not exceed 1 1/2 hours. The relative humidity need not be controlled during the transition periods.

4.5.6.7 Temperature and humidity storage mode). The torpedo control cable assembly, in the packaged condition, shall be subjected to a 28-day temperature/humidity test. The total test is made up of two 14-day temperature and humidity cycles. The term “chamber temperature,” as used in this test, refers to the temperature of the air immediately surrounding the equipment, Chamber temperature shall be changed (1) by varying the temperature of a single chamber, or (2) by moving the test specimen from one constant temperature chamber to another, The fluctuations from all specified temperatures shall not exceed 10 degrees F (5.5°C). The following test sequence is based on a start at 0800 Monday, Regardless of the actual day and time the test is initiated, there shall be no deviation from the test sequence as prescribed,

(a) At 0800 Monday place the test specimen in a chamber which has been maintained at -65 degrees F (-54°C) for at least 2 hours immediately prior to 0800 (humidity not controlled).

(b) At 1600 Monday the chamber temperature shall be changed to +160 degrees F (71°C) and 95 +3 -0 percent humidity. The chamber temperature shall be adjusted to reach 160 degrees F (71°C) and 95 percent humidity not later than 1800.

(c) The test specimens shall be held under these conditions until 0800 Tuesday, at which time the temperature decrease shall begin. The rate of decrease of chamber temperature shall be equal to or greater than 35 degrees F (19°C) per hour for at least 2 1/2 hours. Thus, at 1030, the temperature of the air surrounding the test specimens shall be +70 degrees F (21°C) or lower. The chamber temperature shall reach -65 degrees F (-54°C) no later than 1400 and this temperature shall be held until 1600.

(d) At 1600 Tuesday the chamber temperature shall be changed to +160 degrees F (71°C) and 95 +3 -O percent humidity as rapidly as practicable (not later than 1800) and held until 0800 Wednesday.

(e) On Wednesday, Thursday, and Friday the operations performed on Tuesday shall be repeated.

(f) The chamber temperature shall be raised to +160 degrees F (71°C) and 95 percent humidity on Friday evening and shall be so maintained until 0800 Monday.

(g) At 0800 Monday the sequence of operations described above for Tuesday of the first week shall be followed and shall be repeated daily until Friday of the second week.

(h) On Friday, the chamber temperature shall be reduced to -65 degrees F (-54°C), and this temperature shall be maintained until 0800 Monday of the third week. The cycle will be complete at 0800 Monday.

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4.5,6.7.1 The above sequence of temperature and humidity conditions will constitute one cycle. Two such cycles shall be applied. The second cycle shall be completed at 0800 Monday of the fifth week. At that time, the test specimen shall be allowed to return to room temperature. Without drying, wiping, or baking, the test specimen shall be subjected to tests of 4.5.5.2 and 4.5.5.3.

4.5.6.8 Tropical environment. For this test, the cable assembly shall be mated to Torpedo Tube Cable Connector Mk 42 Mod 1. The coupled unit shall be-mounted in the test chamber wall or suitably protected so as to prevent exposing the in-line connector face and the tube side of the connector.

(a) The test specimen shall be placed in a mold chamber in which a temperature of 86 degrees F ±5 degrees F (30°±3°C) and a relative humidity of 90 percent ±5 percent shall be maintained, The test specimen, while so housed, shall be sprayed with a suspension of mixed spores. The test specimen shall be maintained in this condition for a period of not less than 28 days. At the end of this period, the test specimen shall be allowed to return to room temperature, and it shall be examined for evidence of being nutrient to fungi and shall be subjected to the tests of 4.5.5.2 and 4.5.5.3. If the test specimen shows susceptibility to fungi, color photographs shall be taken of any fungus growth on the test specimen.

(b) One type of fungus from each group listed in MIL-T-18404 shall be used in the test specimen inoculation, In preparing the spore suspension, distilled water (having a pH value between 5,8 and 7.2 at 77 degrees F (25°C)) shall be prepared by sterilization in convenient containers (approximately 100 milliliters (cm^3) each), Approximately 10 milliliters (cm^3) of the solution shall then be introduced directly into a stock culture and shaken vigorously in order that a well sporulated suspension will result without disturbing the agar.

(c) This process shall be repeated for each type fungus, The separate spore suspension from the various types of fungi shall then be mixed together in an atomizer to provide a composite suspension. Actively growing cultures in ripe fruiting condition shall be used for each inoculation and under no condition will the inoculum be kept for more than an 8-hour period.

4.6 Packaging , packing, and marking. Packaging, packing, and marking shall be examined for compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The levels of preservation and packaging shall be as specified (see 6.2),

5.1.1 Level A

5.1.1.1 Unit packaging. The connector shall be protected with a plastic dust cap and be wrapped in a transparent bag conforming to characteristics equal to or better than a 0.003-inch wall thickness of Type 1, Class-E, Style-2 of MIL-B-117. Loose component parts shall be protected from damage and may be enclosed in similar plastic bags, The cable assembly and components shall be packed in 3/4-inch thick cellulosic material conforming to PPP-C-843, Type-II, Class-B and packed in a 12-1/4 x 12-1/4 x 6-13/16 inch fiberboard container in accordance with Drawing 757989.

5.1.2 Level B. Not applicable.

5.1.3 Level C. The cable assembly shall be packaged in accordance with the manufacturer's commercial practice,

5.2 Packing. Packing shall be Level A or Level C, as specified (see 6.2),

5.2.1 Level A.

5.2.1.1 Consolidation container. The cable assemblies, packaged as described in 5.1.1.1, shall be packed, three each, in a wooden box in accordance with PPP-B-621. Class-2 and may be shipped individually in small lots.

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5.2.1.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Pattern No. 91.

5.2.2 Level B. Not applicable,

5.2.3 Level C. The cable assemblies packaged in accordance with 5.1.3 shall be packed to afford protection from damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Marking

5.3.1 Special markings. None, unless otherwise specified.

5.3.2 Normal markings. In addition to the markings required herein or by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6, NOTES

6.1 Intended use. The cable assembly covered by this specification is intended for use aboard submarines in transmitting fire control electrical signals from the fire control console to the torpedo tube door.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.

(b) Mark and Mod of cable assembly required (LD 162531, LD 162533, or DL 2865325).

(c) Assigned activity for preproduction inspection.

(d) Disposition of test samples and failures.

(e) Applicable levels of packaging (see 5.1).

(f) Applicable levels of packing (including palletizing) (see 5.2).

6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data Requirement	Applicable DIDs
4.3.1.2	Certified inspection data	

DIDs (Data Item Descriptions/DI Forms 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DIDs when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

6,3 Metrication. This document has been annotated with metric equivalents of quantities not expressed in the International System of Units. The equivalent units are in accordance with ASTM E380-72 as adopted by the DoD on 29 November 1973, and are included for the purpose of encouraging metrication. Rather than exact equivalents, the metric quantities are generally given in rounded amounts within the tolerances specified herein.

Custodian:
Navy - OS

Preparing Activity:
Navy - OS

Project Number
1355 -N353

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MATERIAL PROCURED UNDER A

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO *(If "Yes", in what way?)*

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